AIRFOIL SELF-NOISE PREDICTOR

Predicts the sound level (decibels) generated by an airfoil based on its physical shape, angle of attack, air speed, and frequency

Team 03 | AI101: Data Science and AI Essentials with Python

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Repo: https://github.com/rajitgupta1267/ai-project-grp-3

PROBLEM AND DATA

- Problem statement

Airfoils generate aerodynamic noise during flight, contributing significantly to environmental noise pollution.

- Core objective

Build a machine learning regression model to accurately predict the sound level (in decibels) generated by an airfoil.

- Why it matters

Wind-tunnel noise tests are slow and costly. A model gives a quick estimate for aircraft, drones, fans and wind turbines.

1,503	5	1
rows	inputs	target (dB)

Column	Meaning
Frequency (Hz)	Pitch of the sound
Angle of attack (deg)	Tilt of the airfoil to the air
Chord length (m)	Size of the airfoil
Air speed (m/s)	Speed of the airflow
Suction side thickness (m)	Shape-related measurement
Sound level (dB)	Target we predict

Source: NASA Airfoil Self-Noise (UCI). This example uses synthetic data with the same columns.

APPROACH

1. Load data : pandas read the csv
2. Check : missing values and ranges
3. Split : 80% train, 20% test
4. Train : linear regression
5. Evaluate :RMSE, MAE, plots etc.

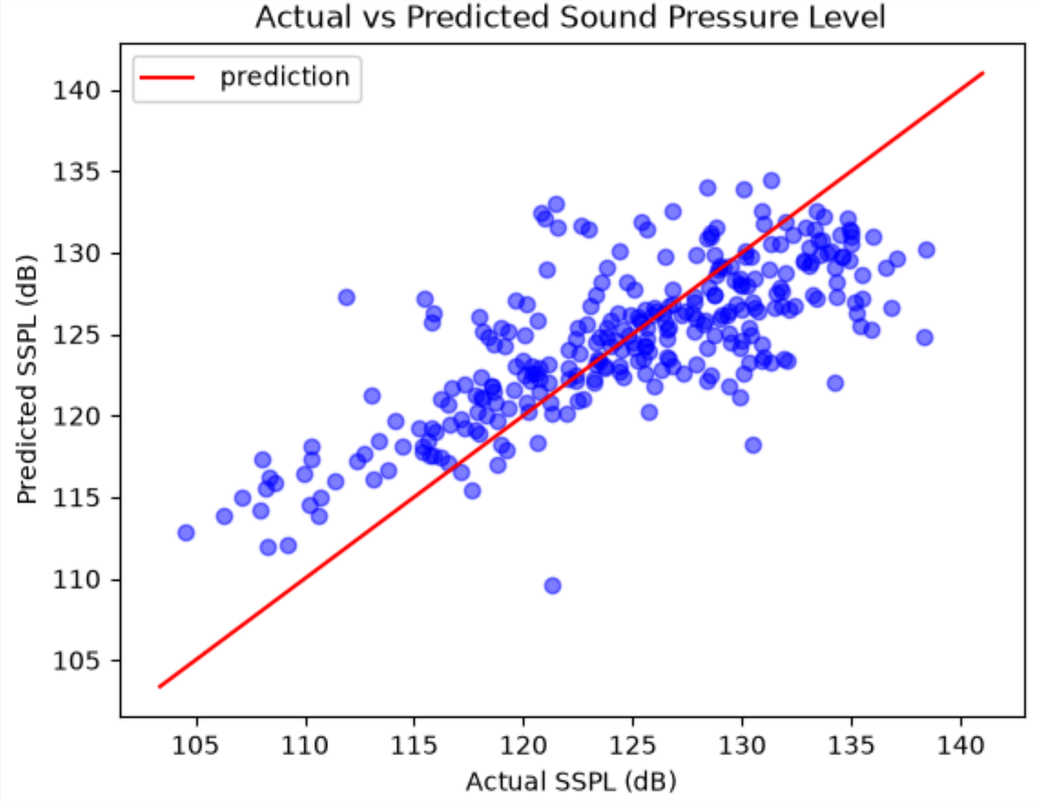
LIBRARIES USED

- pandas
- NumPy
- matplotlib
- scikit-learn

MODEL COMPARED

Linear regression : simple baseline, inputs scaled first

RESULTS



R-squared : 0.558
RMSE (dB) : 4.704
MAE (dB) : 3.672

```
Dataset shape: (1503, 6)
   Frequency  Angle_of_attack  Chord_length  Velocity  Displacement_thickness  \
0        800              0.0        0.3048      71.3                0.002663
1       1000              0.0        0.3048      71.3                0.002663
2       1250              0.0        0.3048      71.3                0.002663
3       1600              0.0        0.3048      71.3                0.002663
4       2000              0.0        0.3048      71.3                0.002663

      SSPL
0  126.201
1  125.201
2  125.951
3  127.591
4  127.461

Coefficients: [-1.0000e-03 -4.0600e-01 -3.4469e+01  9.8000e-02 -1.3946e+02]
Intercept: 132.532
MAE: 3.672
R2 Score: 0.558
RMSE: 4.704
```

LEARNINGS

What worked,What didn’t

- **Worked:** clean dataset (no missing values), end-to-end pipeline runs from one command
- **Worked:** reproducible setup via requirements.txt

- **Didn’t:**No hyperparameter tuning yet
- **Didn’t:**Demo data is synthetic, so results are not real-world

Who did what

- Om Gupta: execution (code and run)

- Vaishali Parmar: slides

- Anish Kumar Rakesh: video

- Rajit Gupta: documentation (README)

AI-Use Disclosure

- Gemini was used to fix an error in src/main.py, provide code to save the plot as .png, and give a first draft of the README (edited further)